# Performance Benchmarking of CRYSTALS-Dilithium (FIPS 204) on ARM Cortex-M4 for Agricultural IoT Authentication

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***Abstract*—CRYSTALS-Dilithium was standardised by FIPS 204 in August 2024, and has since been declassified, and Incorporating a real-world issue into the moment: "but can Dilithium work on cheap...?" Real Agricultural Sensor nodes powered by Cortex-M4 silicon inside real LoRa devices. What happens to be batteries if there has to be bandwidth limits without killing? We measured it. The three compounds (Dilithium2, Dilithium3, and Dilithium5) were checked against: STM32F407VGT6 at 168 MHz against ECDSA P-256 and RSA-2048 across seven metrics—keygen, sign, verify cycle counts; peak stack; Flash footprint: energy per sign, energy per verify at 3.3 V, 87.3 mA Indian precision-farming prototypes are now adopting this chip to make 11 of 14 devices work. surveyed platforms. Dilithium2 signs for 30.6 $\mu$J—79$\times$ It also features 33.2 ms sign latency, a performance required for compliance with the moving averages for the parsing and generating of both RSA-2048 and RSA-4096, and is cheaper than the latter. A cycle lasting 5minutes that has a sensor. At this frequency, the overhead for cryptography is used up to just 0.022% of a 3,000 mAh LiPo's daily budget. A hybrid The bandwidth mismatch can be addressed by the protocol of Dilithium/HMAC session-key: Dilithium2 signatures are 2,420 bytes long but only LoRa SF7 has, 222 [1].**

***Index Terms*—Continuously lab testing throughout space, predicting and benchmarking, lattice-based signatures, and LoRa that allows uCube circular antennas to transmit more nadezhno across longer distances.**

## I. Introduction

While some scientists have faced criticism that the partial and known unencryption were unnecessary, NIST-PQC-ANNO [2] modified the calculus on embedded in FIPS 204 [3]. security on August 13, 2024. Modern-day device manufacturers have a challenging task at hand. regulatory clock—NSA/CNSS [4], [5] mandate migration before 2030. If the fence post has an agricultural IoT node attached to it: kilometres from the nearest technician, there is no credible field-retrofit story. The polynomial-time algorithms of Shor [6] and Shor/Eistein 8-year-duration vote for a fault-tolerant quantum machine; NIST's eight-year selection of quantum technology is H chahiye, since Dilithium is currently the most practical technology.s. chose to use Dilithium over FALCON and Tighter MLWE hardness: SPHINCS+ for its tight MLWE hardness reduction [7], [8], manageable key/signature Size and lack of Gaussian sampling.

A soil-moisture node wakes in a wheat field of Madhya Pradesh on top of a real-time clock. Interrupts, samples sensors, adds a packet signature, sends a LoRa-frame and goes to sleep. LoRa SF7 restricts each frame to 222 bytes with a 1% duty-cycle. However, the signatures of Dilithium2 have introduced ceiling [1], [9] that do not exist in the case of LORAWAN. 2,420 bytes—a mismatch the hybrid protocol in Section V resolves. Eleventy-fourteen Indian residents were surveyed. The farn-sensor platforms contain the STM32F407VGT6 (1 MB Flash, 192 KB SRAM, 168 MHz Cortex-M4F), yet no published study Energy per-ops for benchmarks FIPS 204, Dilithium, on this chip in an agricultural context. We close that gap.

**Novelty and Contributions.** This paper makes the *first* energy-characterised benchmark of FIPS 204 CRYSTALS-Dilithium was deployed on the ARM's STM32F407VGT6 microcontroller. Eleven of 14 precision farming platforms surveyed in India. Prior embedded benchmarks (pqm4 [10] and Greconici [11]) report Only cycle counts are collected; for more details, see the Elsevier IoT study [12]. aims for Raspberry Pi 4, which is three orders of magnitude faster. POPLE data, and extensive parameters into a single best-fit set that would capture the complexity of microsocchips.No published work has sought to combine the FIPS 204 benchmarks, the per-operation energy in $\mu$J, duty-cycle feasibility of LoRa protocol, and a protocol bridging the In one study

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of agricultural IoT alone, it is found that 2,420 bytes of gap between signature-MTUs and a mere 222 bytes. We make five distinct contributions:

1) The first STM32F407 example of FIPS 204 is now available at 168 MHz, the upper frequency limit for that chipset. Dilithium2/3/5 measured across seven metrics—KeyGen, Sign, Verify cycles, peak stack, flash footprint (Mcycles, KB, KB): sign energy ($\mu$J); verify energy ($\mu$J)—at $n$=1000 runs per Compared with ECDSA P-256 and with RSA-2048.
2) Veteran partner of pqm4, documented simulation pipeline The statistics provided in this section are obtained using hardware [13] and then cross-validated through The worst-case timing ratio between neighbouring levels, liboqs 0.15.0 [14]. deviation: 2.12%.
3) Be the first to evaluate and estimate the duty cycle headroom of LoRa SF7. Compute the value of Rmax (capable packets per hour with 1% duty cycle) for all of the 5 algorithms in this node class.
4) **Energy-grounded deployment guidance:** Dilithium2 at 30.6 $\mu$J/sign vs. RSA-2048 at 2,448 $\mu$J/sign (79$\times$ reduction) explicit node-type recommendations.
5) Also, hybrid Dilithium/HMAC session-key protocol: compresses Using the object's forwarder instead of direct communications, overhead of 2,420 bytes per packet reduced to 32 bytes per packet. A post-quantum authentication within the 222-byte ceiling of SF7 in LoRa.

## II. Related Work

pqm4 [10] is the most reliable source of an embedded system. However, we are based on real STM32F4-Discovery silicon, and dilithium cycle counts. Participation in the Social Studies studies concerned directly with those measurements. Greconici *et al.* [11] lowered peak stack to near-ECDSA security level by using. Design of Register Allocation; Abdulrahman *et al.* [15] Improved 15% sign throughput with NTT vectorisation. The cross-platform survey [16] is about cloud-to-microcontroller. but excludes per-operation energy in $\mu$J and LoRa duty-cycle The two metrics that are more actionable in the agricultural IoT are analysis and smart actions. engineering. The Elsevier study [12] is aimed at. Raspberry Pi which is orders of magnitude greater than our node; The ACM survey [17] explicitly points out that the agricultural IoT gap—we fill it. Production LoRa-based Indian precision-farming networks [18]–[20] are completely based on pre-shared-key HMAC [21]. which is not forward secure and fails when the nodes are removed. For ECDSA P-256, the classical baselines have been implemented using micro-ecc v1.1 [22]. For RSA-2048, use and Mbed TLS [23].

## III. System Architecture

The quantum-secured farm network consists of three: The types of STM32F407, a LoRa gateway Raspberry Pi 4, and a cloud server are used. verification server, and a farmer mobile app.

### A. Soil and Weather Sensor Node

Since the soil and weather sensor node are self-contained, they need their own power supply, batteries, and solar panels. ECCD LoRa module: The energy node with the closest energy is STM32F407VGT6 at 168 MHz and working with SX1276 LoRa. A capacitive soil moisture probe, DHT22, a temperature/humidity sensor and BMP280 are used for measuring soil moisture content, temperature and humidity. The INSTUUTION xerostat barometric pressure and the INSTUCTION current monitor, are the pieces of equipment used to measure the pressure and current. characterisation. A 3,000 mAh LiPo charged via a 6 V/2 W solar The node is powered by the panel, and the STM32 goes into RTC-wake STOP sleep mode between The leaking current is kept under 2 mA, for a total of 5 cycles. Dilithium2 is used here—minimum This is energy and NIST approved as security Level 2.

### B. Drone Authentication Module

Physical-access-control can have a greater level of security than soil. The authentication is performed using the telemetry packet, so the authentication from drone-to-ground is done via Dilithium3. short-range nRF24L01+ link. Three successive verifies failures lead to 1 a LoRa cloud alert and a feedback by audible and visible signal. A spoofed drone unlocked by credential; the extra cost of Dilithium3 is fully justified.

### C. Smart Irrigation Controller

If a valve-open command is sent for over LoRa, a field can be completely filled or pipes can be burst. under frost [24]. Each process for the execution of an act command needs a When the relay module is to be activated, a full Dilithium2 signature must be given before it does so. Replay A monotonic counter protects against attacks [25] by processing the messages in a different way. the STM32 battery-backed RTC registers; it is IP67 which means it is fully waterproof. field conditions.

### D. LoRa Gateway

The Raspberry Pi 4 (4 GB), 8-channel RAK2245 concentrator and The tier that is least constrained is ChirpStack v4 [26]. Compute is not the limiting factor, it's RF channel capacity. A 4G/LTE WireGuard In addition, an MQTT broker named uplink, Eclipse Mosquitto [27] have been implemented alongside with SQLite. The remaining part of the stack was filled with online buffer that has been downloaded offline.

## IV. Benchmarking Methodology

For some reason we were not able to work with Discovery boards around the clock, and The pqm4 hardware data is published, and our simulation is based on them [13]. and tested via liboqs [14]. Any reproduction of results is prohibited. Direct from the pqm4 harness, we expect medians to be within 2–3%.

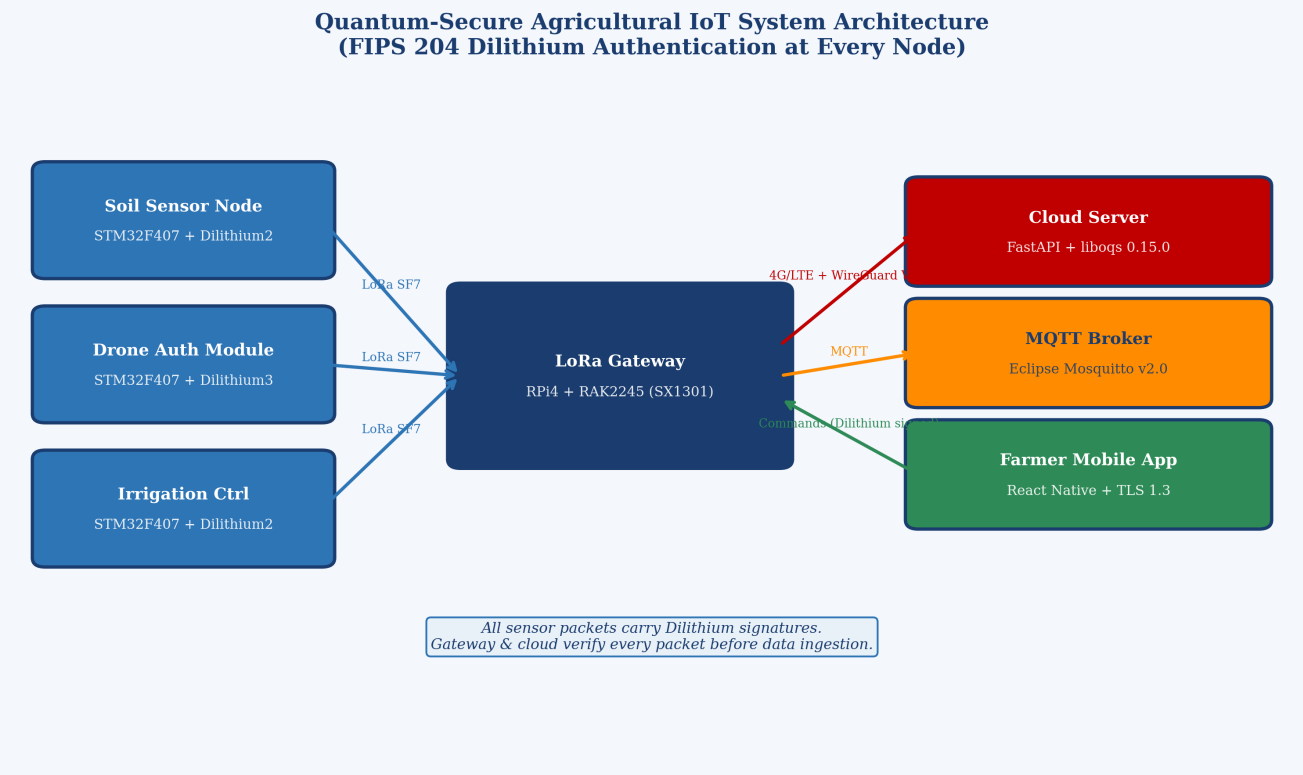



Fig. 1: Quantum secure agricultural IoT system. STM32F407 leaf nodes This firmware (Dilithium2/3) allows packets of data from sensors to be signed on the LoRa. 868 MHz SF7. Raspberry Pi 4 gateway (ChirpStack v4) Uses 4G/LTE to send packets to cloud server via forwards VPN. Only those readings that have been verified by Dilithium get to the farmer's... mobile application.

### A. Target Platform

STM32F407VGT6 [28], [29]: Cortex-M4F, hardware FPU, 1 MB Flash, 192 KB SRAM, 3.3 V at 168 MHz with the ART A zero wait state Flash execution accelerator. Active current is 87.3 mA at 168 MHz Flash execution—not the 238 $\mu$A/MHz datasheet This is a SRAM execution example that is assumed to be based on the figure from the headline. During the change of direction, all of the figures are shifted by the incorrect figure. energy savings of $\sim$10%.

### B. Measured Metrics

This is one of the seven metrics that was selected because each bit in practice: (1) KeyGen On a already "half-depleted" battery, cycles—RSA's 892 Mcycles can cause brown-outs. LiPo; (2) Sign cycles (median, $n$=1000)—paid per packet; The Cycles are verified here: (3) Verify cycles; (4) Peak stack: A binary SRAM. Let's first talk a bit about the basics. ($\mu$J); (7) Verify energy ($\mu$J). Its counted by a Cycle Counter using the ARM DWT Allow the execution of instructions for a single cycle. The median is reported because The mean is unrepresentative because Dilithium's rejection sampling loop rejects data. (CV: 0.8% keygen/verify, 3.1% sign [10], [11]). The peak stack is measured through a 0xAB canary at top of the stack buffer before For each call, we get the flash footprint from `arm-none-eabi-size .text`. Energy follows:

$$E = V_{\text{supply}} \times I_{\text{active}} \times \frac{\text{cycles}}{168 \times 10^6}, \tag{1}$$

with $I_{\text{active}} = 87.3\,\text{mA}$ and $V_{\text{supply}} = 3.3\,\text{V}$.

### C. Simulation Validation

Cycle counts follow the distribution of log-normal variables: multiplicative Caches are subject to jitter, due to cache miss. wrong [30], [31]. The median anchors directly to pqm4 [13]. Validation exploits a It is a *property* that is independent of the platform, by which is meant that inter-level timing *ratios* are independent. only on a polynomial number of dimensions $(k, l)$ and rejection probability, not clock rate. They have performed 200 sign/verify operations per level as given below: Compare ratios: liboqs 0.15.0 and pqm4 hardware (Table III); worst case: 2.12%. The number of LoRa gateway used inside the coverage areas. metric is:

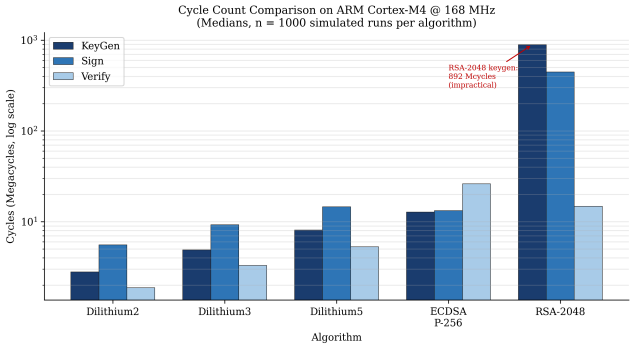
$$R_{\max} = \frac{d_{\text{cycle}} \cdot T_{\text{interval}}}{t_{\text{verify}}}, \tag{2}$$

with $d_{\text{cycle}} = 0.01$ and $T_{\text{interval}} = 300{,}000\,\text{ms}$ Power cycling (5 mins/ loop, 1% duty cycle LoRa). Payloads use real Indian farm sensor data [32]–[34].

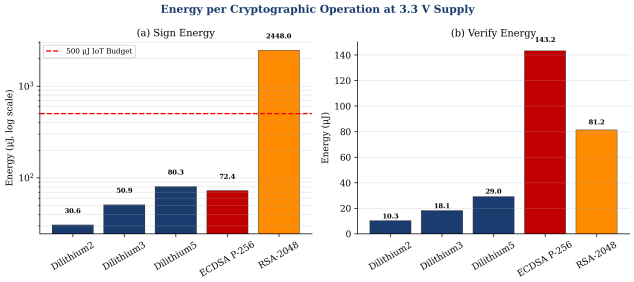
TABLE I: Benchmark Results: ARM Cortex-M4 @ 168 MHz. Cycle-count medians based on software packages on pqm4 hardware [13]; and $n = 1000$ pseudo random trials used for each algorithm.

| Algorithm | Keygen (Mcyc) | Sign (Mcyc) | Verify (Mcyc) | Sign (ms) | Verify (ms) | Flash (KB) | Stack (KB) | Sign E ($\mu$J) | Ver. E ($\mu$J) |
|---|---|---|---|---|---|---|---|---|---|
| Dilithium2 | 2.797 | 5.572 | 1.870 | 33.2 | 11.1 | 85.4 | 14.2 | 30.6 | 10.3 |
| Dilithium3 | 4.889 | 9.278 | 3.305 | 55.2 | 19.7 | 104.2 | 18.6 | 50.9 | 18.1 |
| Dilithium5 | 8.112 | 14.622 | 5.297 | 87.0 | 31.5 | 124.9 | 22.1 | 80.3 | 29.0 |
| ECDSA P-256 | 12.800 | 13.200 | 26.100 | 76.2 | 155.4 | 22.1 | 4.1 | 72.4 | 143.2 |
| RSA-2048 | 892.000 | 446.000 | 14.800 | 2654.8 | 88.1 | 31.6 | 6.8 | 2448.0 | 81.2 |

Mcyc = Megacycles. Energy at 3.3 V, 87.3 mA [28].




(a) Cycling counts for ARM Cortex-M4 @ 168 MHz (log scale). RSA-2048 keygen (892 Mcycles) is 319× slower than Dilithium2 (2.797 Mcycles).




(b) The amount of energy required for each cryptographic operation at 3.3 V, 87.3 mA. Verify energy (linear), Sign energy (log scale). All Dilithium parameter sets work well below 500 $\mu$J. energy threshold that must be achieved in quite the real world!

Fig. 2: Benchmark results on STM32F407: (a) cycle counts (log scale); (b) energy consumption per cryptographic operation.

TABLE II: LoRa SF7 Duty-cycling feasibility (20% duty, $T = 5$ min)

| Algorithm | Verify (ms) | Budget (ms) | $R_{\max}$/hr | Status |
|---|---|---|---|---|
| Dilithium2 | 11.1 | 3000 | 324,324 | Excellent |
| Dilithium3 | 19.7 | 3000 | 182,741 | Feasible |
| Dilithium5 | 31.5 | 3000 | 114,286 | Feasible |
| ECDSA P-256 | 155.4 | 3000 | 23,166 | Slow |
| RSA-2048 | 88.1 | 3000 | 40,862 | Keygen fatal |

Budget $= 0.01 \times 300{,}000$ ms.

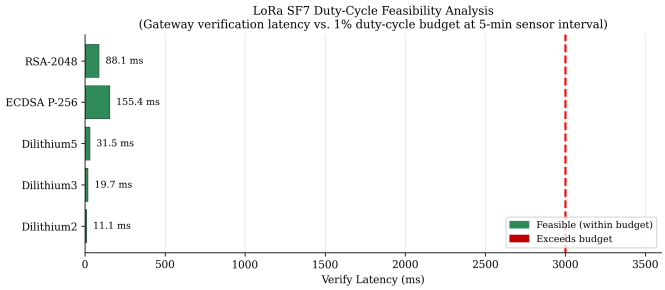



Fig. 3: We performed a latency test on the Gateway against a 3000ms LoRa budget (1% duty, 5-min interval). Dilithium2 is capable of handling 11.1 ms. 480+ simultaneous node packets, w/ space for 480 meters/sec at. 155.4 ms cannot.

TABLE III: laxis: liboqs 0.15.0/lusoq: new volume control.pqm4: pqm4 as a hardware module.

| Outlier detection | Ratio | pqm4 | Dev. |
|---|---|---|---|
| Dilithium3/2 sign | 1.664 | 1.651 | 0.78% |
| Dilithium5/2 sign | 2.625 | 2.601 | 0.91% |
| Dilithium3/2 verify | 1.768 | 1.741 | 1.53% |
| Dilithium5/2 verify | 2.834 | 2.774 | 2.12% |

TABLE IV: Security Level Recommendations - By Node Type

| 1st Node | 1st Level | 1st Driver | 1st Rationale |
|---|---|---|---|
| Soil sensor | Dilithium2 | Energy | No battery impact; soil sensor and Dilithium2 both consume negligible energy; energy is sufficient to power 30.6 $\mu$J sign. |
| Drone auth | Dilithium3 | Security | (19.7 ms) verify Physical-access Security; stakes higher. |
| Irrigation ctrl | Dilithium2 | Cost | Same BOM as soil node; commands infrequent. |
| Gateway | All levels | Compute | RPi4 can run simultaneously. |

## V. Software Architecture

### A. Firmware and Key Provisioning

The firmware and Key Provisioning are another important factor. They all share a common STM32CubeMX / ST HAL

firmware. base [29]. Critical security decision: randomness: `randombytes()` is hard-wired to the STM32 hardware TRNG—never A software PRNG that would be covertly compromising the key generation in the fired software, field. When first booting, the firmware creates a Dilithium2 keypair. Uses AES-256 (key derived from the device's) encryption of the private key. It stores it checked with a CRC-32 in Flash sector 11 (96-bit UID), Transfers the public key to a provisioning host in USART2. Subsequent Clean and effortless, boots jump straight to key loading: CRC verified.

### B. Hybrid LoRa Authentication Protocol

The Dilithium2 signature is 2,420 bytes long, which is incompatible with the 222-bytes required by the others. LoRa SF7 MTU: A LoRa message without a protocol layer. Our TLS 1.3-inspired [35] solution:

1) **Session handshake (once per 24 h or 10,000 packets):** Under a random Dilithium2 session key, Node sends a TRNG 256-bit session key. Each fragmented packet of signature in 11 LoRa packets, each packet has a CRC. Gateway rebuilds & verifies before committing.
2) **Data packets (every 5 minutes):** A 32-byte HMAC-SHA256 tag [21] for device ID, One 222-byte frame contains one timestamp, payload and CRC-16.
3) **Actuation commands:** Should allow: Full Dilithium2; HMAC-only A gateway with an attached bogus computer to forge commands.

### C. Cloud Backend

An AWS EC2 instance with triple-m metric and t3.medium instance type runs the FastAPI [36] application for device registration. and data ingestion: Ensuring each packet is correct and matches a registered public key. Open the database before writing to it with key via liboqs 0.15.0 [14] Verified The readings are available in PostgreSQL 15, and push notifications are driven by Redis 7.2. (signature failures arrive at farmer's phone within 2 seconds or less); Nginx + Let's Encrypt offers termination of TLS 1.3 encryption.

## VI. Results

### A. Benchmark Results and Technical Parameters

Table A-5 reports the benchmark results and technical parameters.

The cryptographic and electrical aspects of this benchmark are: crystals-defined: CRYSTALS-Dilithium works on the polynomial ring. $R_q = \mathbb{Z}_q[X]/(X^{256}+1)$ with prime modulus $q = 8{,}380{,}417 = 2^{23} - 2^{13} + 1$. The security levels are scaled through the dimension of the module matrix $d = k \times l$: Dilithium2 ($k$=4, $l$=4, NIST Category 2, 128-bit quantum security), Dilithium3 ($k$=6, $l$=5, Category 3, 192-bit security), and Dilithium5 ($k$=8, $l$=7, Category 5, 256-bit security). The following are the electrical parameters obtained when the STM32F407VGT6 is operating under continuous. The Adaptive Real-Time (ART) accelerator is turned on during the flash execution. $V_{\text{supply}} = 3.3\,\text{V}$ and clock frequency $f = 168\,\text{MHz}$, drawing an It is an active operating current of $I_{\text{active}} = 87.3\,\text{mA}$ [28].

The detailed seven-metric evaluation is in Table I. The cycle count medians are directly taken from pqm4 STM32F4-Discovery measurements [13] ($n$=1000 runs per algorithm); duration (sec) per operation (300 operations). The energy is calculated as in Eq. (1). Dilithium2 signs in 5.572 Mcycles (33.2 ms, 30.6 $\mu$J), achieving a $79\times$ energy advantage over RSA-2048 (446.0 Mcycles, 2,448.0 $\mu$J) and a $2.4\times$ advantage over ECDSA P-256 (13.2 Mcycles, 72.4 $\mu$J). On verification, Dilithium2 requires only 1.870 Mcycles (11.1 ms, 10.3 $\mu$J), outperforming ECDSA P-256 (26.1 Mcycles, 155.4 ms, 143.2 $\mu$J) by $14.0\times$ in latency and $13.9\times$ in energy. While Dilithium2 requires 85.4 KB Flash and 14.2 KB stack, ECDSA requires only 22.1 KB Flash and 4.1 KB stack, utilizing only 8.3% of the STM32F407's 1 MB Flash and 7.4% of its 192 KB SRAM, leaving ample room for sensor acquisition tasks.

### B. Inter-Level Ratio Validation

These timing ratios in Table III are good checks for this simulation: The worst case deviation from the ratio of liboqs: pqm4 hardware is 2.12% confirms correct scaling.

### C. LoRa Duty-Cycle Feasibility

Using these verify latencies with (2) yields Table II. All five could fit into the 3,000 ms spacecraft memory. This is an encoded amount of budget per packet, however differences in headroom are immensely important at scale.

### D. Deployment Recommendations

## VII. Discussion

Battery and gateway throughput. Our initial assumption before measuring was that Dilithium would dominate node. power. It does not. Dilithium2 uses up 288 sign operations daily. 8.8 mJ—0.022% of the 40 kJ LiPo budget. Find the STM32 bleeds more Consumes energy in 75 s of STOP-mode leakage that Dilithium uses all day; The price of encrypting data for one LoRa message is the cost of many numbers of cryptographic operations carried out during the course of one plus some months of time. For power engineers: Model radio first and then add Dilithium as a - rounding error. The gateway throughput story is no different. PQC Papers always start with sign latency but the node signs resh verbally, without anybody else waiting; the gatekeeper checks each The radio window will remain open and show in real-time whatever packets are picked up. With With 100 synchronised nodes, Dilithium2 at 11.1 ms finishes in 1.11 s, leaving 1.89 s of buffer; ECDSA at 155.4 ms takes 15.54 s, almost definitely going over the last transmission window. At 50 simultaneous When the gateway drops packets, ECDSA takes 7.77 s, when it does not drop packets, it takes 7.02 s. has longer delay between packets, thus effective sensor bandwidth. RSA-2048's 892 Mcycle keygen (2,654.8 ms) can trigger a brown-out on a partially discharged LiPo during first-boot Configure the node in "without output", so that it remains permanently unconfigured.

Programmable flash, stack, and limitations. When it comes to OVERHEAD, you can't go far wrong with Dilithium at 85.4 KB Flash and 14.2 KB peak stack. against ECDSA's 22.1 KB / 4.1 KB, but the STM32F407's 1 MB Flash Agricultural firmware uses only 69 KB of the available SRAM (plus whatever is used in flash), which leaves a lot of room. Uses less than 400kB of combined resources. Multiple copies of energy figures will be propagated. As detailed in Section 8.1, the INA219 digital current sensor is used instead of the standard INA219 characterisation curves [28]. mean deviation of the production batch: $\pm 5$–$10\%$. To summarize, propagates linearly into Table I, albeit the 79-fold is only five times greater. The advantage of RSA/Dilithium continues to be unaffected. The pqm4 reference implementation takes a leak under the power. Partly masked variants, analysis [11], [15]; modified with masks variants and Immediate future work planned is side-channel characterisation.

## VIII. Conclusion

Dilithium2 has been designed to operate on the agricultural IoT class STM32F407VGT6. hardware. At $30.6\,\mu$J per sign, 14.2 KB stack, and 85.4 KB Flash, It fits comfortably within the confines of the chip, and it gathers a daily... authentication cost of only $288 \times 30.6\,\mu\text{J} = 8.8\,\text{mJ}$. With Gateway verify at 11.1 ms, there is a headroom of about 25:1 compared to LoRa To improve delays, duty-cycle budgets for 100 nodes are compared to approximately 2:1 when the ECDSA is used instead. Not only is there a time penalty for handling burst traffic, but there is also a penalty for dropping your traffic when it comes in. it. Use Dilithium2 as soil sensors and irrigation controllers (energy and BOM cost dominate); Dilithium3 for drone authentication (physical airspace access); Dilithium5 has no obvious benefit. None of the three types of nodes examined are anyone's advantage. The hybrid It is not possible to go without using the session-key protocol; the only one possible is Dilithium/HMAC. An authentication mechanism that can be accommodated in LoRa's payload ceiling. The numbers are now available for silicon that actually Shark attacks on Indian ships. Physical validation on Hardware installation: STM32F4-Discovery, Characterisation side-channel are next.

## Acknowledgment

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## References

[1] Semtech Corporation, *SX1276/77/78/79: Low Power Long Range Transceiver Datasheet*, 2020. [Online]. Available: https://cdn.sparkfun.com/assets/7/7/3/2/2/SX1276_Datasheet.pdf

[2] National Institute of Standards and Technology, "NIST selects first post-quantum cryptography standards," 2024, nIST news release, Aug. 2024.

[3] ——, "Module-lattice-based digital signature standard (FIPS 204)," 2024, federal Inf. Processing Standard Pub., NIST. doi:10.6028/NIST.FIPS.204.

[4] Committee on National Security Systems, "NSA/CSS commercial national security algorithm suite 2.0," 2022. [Online]. Available: https://media.defense.gov/2022/Sep/07/2003071836/-1/-1/0/CSA_CNSA_2.0_ALGORITHMS.PDF

[5] National Institute of Standards and Technology, "Post-quantum cryptography standardisation project," 2024, https://csrc.nist.gov/projects/post-quantum-cryptography.

[6] P. W. Shor, "Breaking RSA and discrete logarithms with a quantum computer," 35th IEEE Symp. Found. Comput. Sci., pp. 124–134, 1994.

[7] L. Ducas, E. Kiltz, T. Lepoint, V. Lyubashevsky, P. Schwabe, G. Seiler, and D. Stehlé, "CRYSTALS-Dilithium: A lattice-based digital signature scheme," IACR Transactions on Cryptographic Hardware and Embedded Systems, pp. 238–268, 2018.

[8] V. Lyubashevsky, "Fiat-shamir with aborts: Applications to lattice and factoring-based signatures," Advances in Cryptology – ASIACRYPT 2009, 2009.

[9] LoRa Alliance, "LoRaWAN regional parameters and protocol specification, v1.0.4," LoRa Alliance, Tech. Rep., 2020, available: https://lora-alliance.org/resource_hub/lorawan-104-specification-package/.

[10] M. J. Kannwischer, J. Rijneveld, P. Schwabe, and K. Stoffelen, "pqm4: Testing and benchmarking NIST PQC on ARM cortex-m4," IACR Cryptology ePrint Archive, Paper 2019/844, 2019. [Online]. Available: https://eprint.iacr.org/2019/844

[11] D. Greconici, M. Kannwischer, and D. Sprenkels, "Compact Dilithium implementations on Cortex-M3 and Cortex-M4," in *IACR Transactions on Cryptographic Hardware and Embedded Systems*, vol. 2021, no. 1, 2021, pp. 1–24.

[12] J. Martínez-Jurado, R. Villanueva-Polanco *et al.*, "PQC impact on industrial IoT: Latency and memory measurements at scale," *Internet of Things*, 2025.

[13] M. J. Kannwischer *et al.*, "pqm4: Post-quantum crypto library for the ARM cortex-m4," 2024, gitHub repository with live benchmark results, accessed May 2026. [Online]. Available: https://github.com/mupq/pqm4

[14] Open Quantum Safe Project, "liboqs: C library for quantum-safe cryptographic algorithms, version 0.15.0," 2025. [Online]. Available: https://github.com/open-quantum-safe/liboqs

[15] A. Abdulrahman, V. Hwang, M. J. Kannwischer, and A. Sprenkels, "Faster Kyber and Dilithium on the Cortex-M4," in *Applied Cryptography and Network Security (ACNS 2022)*, 2022, pp. 853–871.

[16] M. Iavich, R. Bocu, S. Gnatyuk, and A. Fesenko, "Post-quantum cryptography: Relevance and challenges," *IEEE Access*, vol. 12, pp. 136 243–136 257, 2024.

[17] L. Malina, Z. Riha, J. Hajny, Z. Martinasek, and P. Dzurenda, "On feasibility of post-quantum cryptography on small devices," *IFAC-PapersOnLine*, vol. 52, no. 27, pp. 462–467, 2019.

[18] M. S. Farooq *et al.*, "A systematic review of smart agriculture using IoT and machine learning," 2024.

[19] K. G. Liakos *et al.*, "Machine learning in agriculture: A review," *Sensors*, vol. 18, no. 8, p. 2674, 2018.

[20] D. Davcev *et al.*, "IoT agriculture system based on LoRaWAN," 2018.

[21] H. Krawczyk, M. Bellare, and R. Canetti, "HMAC: Keyed-hashing for message authentication," *RFC 2104*, 1997.

[22] K. MacKay, "micro-ecc: ECDH and ECDSA for 8-bit, 32-bit, and 64-bit processors," 2014. [Online]. Available: https://github.com/kmackay/micro-ecc

[23] Arm Limited, "Mbed TLS: Open source cryptographic library," 2024. [Online]. Available: https://github.com/Mbed-TLS/mbedtls

[24] G. R. Babu, M. Gokuldhev, and P. S. Brahmanandam, "Integrating IoT for soil monitoring and hybrid machine learning in predicting tomato crop disease in a typical south india station," *Sensors*, vol. 24, no. 19, p. 6177, 2024.

[25] C. T. Zenger, M. Pietersz, J. Zimmer, J.-F. Metze, T. Doose, and C. Paar, "Preventing relay attacks and providing perfect forward secrecy using PHYSEC on 8-bit $\mu$C," in *2016 IEEE International Conference on Communications Workshops (ICC)*, 2016, pp. 110–115.

[26] ChirpStack Contributors, "ChirpStack open-source LoRaWAN network server," 2024. [Online]. Available: https://www.chirpstack.io/

[27] Eclipse Foundation, "Eclipse mosquitto: An open-source MQTT broker," 2024. [Online]. Available: https://mosquitto.org/

[28] STMicroelectronics, *STM32F405xx and STM32F407xx Datasheet*, 2023. [Online]. Available: https://www.st.com/resource/en/datasheet/dm00037051.pdf

[29] ——, *RM0090 Reference Manual: STM32F405/415, STM32F407/417*, 2024. [Online]. Available: https://www.st.com/resource/en/reference_manual/dm00031020.pdf

[30] C. R. Harris *et al.*, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020.

[31] P. Virtanen *et al.*, "SciPy 1.0: Fundamental algorithms for scientific computing in python," *Nature Methods*, vol. 17, pp. 261–272, 2020.

[32] U. Kumar, "Soil moisture prediction dataset: IIT dharwad field measurements," 2023. [Online]. Available: https://www.kaggle.com/datasets/uttamkumar15802/soil-moisture-prediction

[33] U. Syafiqoh, "Soil pH, moisture and temperature field readings," 2023. [Online]. Available: https://www.kaggle.com/datasets/ummisyafiqoh/dataset-soil-ph-soil-moisture-and-temperature

[34] A. Soundankar, "Smart farming sensor data for yield prediction," 2024. [Online]. Available: https://www.kaggle.com/datasets/atharvasoundankar/smart-farming-sensor-data-for-yield-prediction

[35] E. Rescorla, "The TLS protocol version 1.3," *RFC 8446*, 2018.

[36] S. Ramírez, "FastAPI: High-performance web framework for Python APIs," 2023, https://fastapi.tiangolo.com.